Capacity Portfolio Evaluation Under Multi-Dimensional Uncertainty: A Stochastic Simulation Study

Omar Mourssi — 1008311961

April 15, 2026

1 Introduction

Long-term electricity capacity planning requires selecting a resource mix that balances cost and reliability across plausible future conditions. In Ontario, the IESO projects a 75% increase in provincial electricity demand by 2050, driven by electrification, electric-vehicle supply chains, data centres, and industrial expansion [3]. As the generation mix shifts toward variable renewables and battery storage, deterministic planning methods that evaluate a small number of fixed scenarios may not reveal when uncertainty changes which resource mix performs best. This study evaluates 16 candidate portfolios under multi-dimensional uncertainty using chronological Monte Carlo simulation calibrated to Ontario empirical data. The preferred mix under stochastic evaluation (P04, 70/20/10) differs from the deterministic recommendation (P03, 60/30/10) by $3,968M per year. This Deterministic Benchmark Gap is robust across a 2^3 uncertainty factorial, three VOLL values, a UCAP-normalized re-evaluation, and a stochastic adequacy search.

Recent literature emphasizes explicit uncertainty treatment in capacity planning. Roald et al. [7] review optimization under uncertainty in power systems and frame long-term planning as a decision problem under layered uncertainty. Stover et al. [8] argue that deterministic reserve-margin rules are insufficient for high-renewable systems, and Leibowicz et al. [5] show that capturing operational detail in Monte Carlo adequacy simulation changes which resources are identified as adequate. Wang et al. [9] show that temporal structure matters when evaluating the adequacy contribution of renewables and storage, which directly motivates the NORTA AR(1) input model used in this study. A particularly relevant contribution is presented by Park and Baldick [6], who formulate a multi-year stochastic capacity expansion model using scenario-tree discretization of demand and wind, and show that uncertainty changes the optimal build timing and resource mix. The present study adapts that uncertainty-aware logic to a different problem. Whereas Park and Baldick jointly optimize build decisions against a scenario tree, this study fixes the portfolio set and applies simulation-based *ranking and selection* to the evaluation subproblem. Scenario-tree enumeration is replaced by 200 chronological CRN replications at 8,760-hour resolution, and the

stochastic program is replaced by the Kim–Nelson (KN) procedure [4] with Common Random Numbers (CRN) and Bonferroni-corrected confidence intervals. The shared conclusion is the same in both settings: deterministic planning undervalues firm capacity when renewable shortfalls and outages are temporally correlated.

Candidate portfolios are generated by discretizing the feasible allocation simplex over firm generation, renewable generation, and battery storage at 10% increments subject to a fixed total capacity budget. The simulation model is a stylized one-zone system, not calibrated to a specific jurisdiction, with input distributions calibrated from Ontario empirical data (IESO demand records and the CODERS dataset [2]).

The remainder of this report is organized as follows. Section 2 defines the problem and performance measure. Section 3 describes the simulation model. Section 4 presents the experimental design. Section 5 reports the results. Section 6 discusses limitations and future work.

2 Problem Description

2.1 Portfolio Generation

The system comprises three technology classes: firm generation (nuclear, flexible firm, mid-merit gas, gas peakers), variable renewables (wind, solar), and battery storage. Total installed capacity is fixed at $C_{\text{total}} = 12{,}000$ MW, derived from 8,000 MW peak demand and a 50% reserve margin. Each portfolio P_i is defined by $(f_i^{\text{firm}}, f_i^{\text{renew}}, f_i^{\text{stor}})$ with $f_i^{\text{firm}} + f_i^{\text{renew}} + f_i^{\text{stor}} = 1$.

The feasible simplex is discretized at 10% increments subject to $f^{\text{firm}} \in [0.20, 0.70]$, $f^{\text{renew}} \in [0.10, 0.60]$, $f^{\text{stor}} \in [0.10, 0.30]$, yielding $k = 16$ portfolios (Appendix B). This systematic approach avoids selection bias and ensures uniform coverage of the design space.

2.2 Performance Measure

For portfolio P_i in replication j:

$$Y_{ij} = F_i + C_{ij} + \lambda U_{ij}, \tag{1}$$

where F_i is the annualized fixed cost (deterministic), C_{ij} is simulated annual operating cost, U_{ij} is unserved energy (MWh), and $\lambda = \$10{,}000/\text{MWh}$ is the value of lost load. Operating cost is computed from hourly dispatch:

$$C_{ij} = \sum_{t=1}^{8760} \sum_{g \in \mathcal{G}} c_g \, x_{g,i,j,t}, \tag{2}$$

where c_g is marginal cost and $x_{g,i,j,t}$ is dispatched output in hour t. Both C_{ij} and U_{ij} emerge from the simulation described in Section 3; they are not sampled directly. The objective is $i^* = \arg\min_i \theta_i$ where $\theta_i = \mathbb{E}[Y_i]$.

2.3 Research Questions

1. Which portfolio minimizes expected total cost under uncertainty, and does this differ from the deterministic recommendation?
2. Which uncertainty factors materially affect portfolio ranking?

3 Model Description

3.1 System Architecture

Each portfolio allocates capacity across four firm sub-types, two renewable technologies, and battery storage. Firm generators are dispatched in merit order (Table 1); renewables have zero marginal cost; storage follows a rule-based heuristic. Full parameter values are in Appendix A.

Table 1: Firm generator merit-order stack.

Sub-type	Share	MC ($/MWh)	FOR	Capital ($/kW-yr)
Nuclear	20%	22	5%	400
Flexible firm	35%	35	6%	100
Mid-merit gas	30%	50	5%	70
Gas peaker	15%	80	8%	70

Nuclear marginal cost of $22/MWh reflects the variable operating cost of refurbished Ontario nuclear units (fuel plus variable O&M), not the all-in levelized cost of electricity.

3.2 Hourly Dispatch Algorithm

Each replication simulates 8,760 hours of system operation for a given portfolio. The dispatch model uses a sequential priority structure: renewable output is absorbed first (zero marginal cost), storage responds to net-load conditions, and firm generators are dispatched in ascending marginal-cost order to serve any remaining demand. This merit-order approach reflects standard practice in power system operations, where the cheapest available generation is dispatched first to minimize operating cost. The model does not co-optimize storage and firm dispatch; this simplification keeps the project tractable while preserving the core cost and adequacy dynamics that differentiate portfolios.

For each hour t, the dispatch proceeds as follows:

1. **Demand.** $D(t) = D_{\text{base}}(t)(1 + \sigma_d \varepsilon(t))$, $\varepsilon(t) \sim N(0, 1)$, $\sigma_d = 0.09$. Multiplicative noise lets absolute deviations scale with load; the iid assumption is flagged in Section 6.
2. **Renewables.** Wind and solar capacity factors are drawn from NORTA AR(1) Beta distributions (Section 3.3). Aggregate renewable output is $R(t) = \text{MW}_w \cdot \text{CF}_w(t) + \text{MW}_s \cdot \text{CF}_s(t)$.

3. **Residual demand.** $D_{\text{res}}(t) = \max(0,\ D(t) - R(t))$. This is the load that must be served by storage and firm generation.

4. **Storage.** The battery discharges when residual demand exceeds the 70th percentile of the base demand profile and charges from surplus when residual demand falls below the 30th percentile, subject to 85% round-trip efficiency and state-of-charge, power, and remaining-demand limits. These are stylized threshold rules, not co-optimized dispatch; the simplification is noted in Section 6.

5. **Firm dispatch.** Available firm generators are dispatched in merit order (Table 1). For each sub-type g, dispatched output is $x_g(t) = \min(\text{cap}_g \cdot a_g(t),\ D_{\text{rem}}(t))$, where $a_g(t)$ is the outage-adjusted availability fraction. The algorithm walks up the merit-order stack, accumulating operating cost as $x_g(t) \cdot c_g$ and subtracting dispatched output from remaining demand.

6. **Unserved energy.** $u(t) = \max(0,\ D_{\text{rem}}(t))$ after all dispatch. Any demand that cannot be served after exhausting renewables, storage, and all available firm generation is recorded as unserved energy for that hour.

Forced outage modeling. Each firm sub-type is represented by 10 discrete units of equal size. For each sub-type on each day, each unit independently draws a Bernoulli$(1 - \text{FOR})$ availability indicator, producing discrete availability steps of $0\%, 10\%, \ldots, 100\%$. Outage states persist for 24 hours. This per-unit, per-day granularity is coarser than a continuous-time Markov model but captures the daily availability risk that drives adequacy outcomes.

Annual outputs are accumulated over all 8,760 hours: $C_{ij} = \sum_t \sum_g c_g x_{g,t}$, $U_{ij} = \sum_t u(t)$, and $Y_{ij} = F_i + C_{ij} + \lambda U_{ij}$.

3.3 Input Models and Empirical Calibration

Input distributions are calibrated from IESO hourly Ontario demand [3] and CODERS hourly capacity factors for 2,984 Ontario generation sites [2].

Demand. The base profile $D_{\text{base}}(t)$ is fitted via OLS to 2024 IESO data using two seasonal harmonics (annual and semi-annual, capturing Ontario's winter/summer demand peaks) plus a diurnal harmonic, achieving $R^2 = 0.59$. The noise parameter $\sigma_d = 0.09$ is the standard deviation of relative residuals $(D_{\text{obs}} - D_{\text{base}})/D_{\text{base}}$.

Wind. Hourly aggregate CFs are fitted to Beta$(6.73, 15.69)$ via MLE (KS $= 0.021$, mean $= 0.300$). To preserve the strong temporal persistence (empirical lag-1 ACF $\hat{\rho} = 0.995$), CFs are generated via NORTA with AR(1) dependence:

$$Z(t) = \rho Z(t-1) + \sqrt{1-\rho^2}\,\epsilon(t), \quad \text{CF}(t) = F^{-1}_{\text{Beta}}(\Phi(Z(t))). \tag{3}$$

This preserves the Beta marginal while inducing hourly autocorrelation. The distinction from iid sampling is consequential: iid draws overestimate hourly variability while underestimating sustained low-output periods that drive adequacy risk (Appendix D).

Solar. $\mathrm{CF}_s(t) = E(t) \cdot K(t)$, where $E(t)$ is a sinusoidal diurnal envelope (hours 7–21) and $K(t) \sim \mathrm{Beta}(1.77, 6.91)$ via NORTA AR(1) with $\rho = 0.928$, calibrated from CODERS.

Forced outages. See Section 3: 10 units per sub-type, independent daily Bernoulli$(1 - \mathrm{FOR})$ draws.

4 Experimental Design

Each simulation run is a *terminating simulation* over 8,760 hours (one operating year), so output analysis uses independent replications rather than batch means. Error is controlled at three levels. First, the Kim–Nelson ranking-and-selection procedure guarantees $P(\mathrm{CS}) \geq 0.95$ for the indifference-zone parameter $\delta = \$50\mathrm{M}$. Second, the top-4 evaluation uses Bonferroni-corrected pairwise 95% confidence intervals with $\alpha_{\mathrm{pair}} = 0.05/6 = 0.0083$. Third, relative half-widths are reported for every mean (Table 2).

4.1 Common Random Numbers

All portfolios share identical random inputs within each replication, implemented via NumPy's `SeedSequence` spawning. Three independent streams (demand, outages, renewables) are deterministically generated from a base seed and replication index. As a variance-reduction technique, CRN induces positive covariance between outputs and thereby reduces $\mathrm{Var}(Y_i - Y_\ell)$, which is the quantity that drives pairwise inference in ranking and selection. Empirically, variance reduction on pairwise differences ranges from 2× to 35× (Appendix, Figure 9).

4.2 Kim–Nelson Ranking and Selection

The Kim–Nelson (KN) sequential ranking-and-selection procedure [4] selects the portfolio with smallest θ_i with probability of correct selection $P(\mathrm{CS}) \geq 1 - \alpha$, provided the best system is at least δ better than the second best (the indifference-zone assumption). Parameters: $\alpha = 0.05$, indifference zone $\delta = \$50\mathrm{M}$ (roughly 1% of annual system cost, below which a planner would not distinguish portfolios), $n_0 = 30$. The procedure initializes with n_0 CRN replications from all $k = 16$ portfolios, computes pairwise difference variances $S^2_{i\ell}$ (which automatically capture CRN covariance), then sequentially eliminates portfolios whose running means exceed a competitor's beyond a shrinking continuation region. The procedure terminates when one survivor remains.

4.3 Deterministic Benchmark Gap

The deterministic recommendation P_{DET} is obtained by setting all inputs to expected values and selecting the lowest-Y portfolio. P_{DET} and P_{KN} are then evaluated for 200 paired CRN replications. The DBG $= \bar{Y}_{\text{DET}} - \bar{Y}_{\text{KN}}$ quantifies the cost of ignoring uncertainty in portfolio selection. This is not the classical VSS (which compares stochastic and deterministic optimizations); since we select from fixed portfolios, the DBG measures the value of uncertainty-aware selection.

4.4 Sensitivity Analysis

A 2^3 factorial tests demand noise ($\sigma_d \in \{0.03, 0.15\}$), forced outage rates (all 1% vs. realistic), and renewable variability (fixed mean vs. stochastic). The design is restricted to the top-4 portfolios, with 30 CRN replications per cell.

4.5 Extensions

Three additional analyses extend the core evaluation. First, a *stochastic adequacy search* finds, for each top-4 mix, the minimum total capacity C^* such that $\mathbb{E}[\text{EUE}(C)] \leq 10$ MWh/year. This is a one-dimensional simulation-optimization problem (stochastic root finding for $\mathbb{E}[\text{EUE}(C)] - 10 = 0$): at each candidate C, 50 CRN replications estimate the noisy objective, and bisection iterates until the threshold is crossed within tolerance. Second, a *UCAP-normalized comparison* re-evaluates six portfolios (spanning 20–70% firm) each sized to equal effective capacity (8,000 MW UCAP) using ELCC values from IESO planning conventions (Appendix C); this tests whether the equal-installed-MW design biases results toward firm-heavy portfolios. Third, a *post-processing VOLL sensitivity* recomputes $Y = F + C + \lambda U$ over $\lambda \in \{5{,}000, 10{,}000, 15{,}000\}$ using the replication-level components from the extended CRN run.

5 Results

5.1 Portfolio Selection

The KN procedure selects P04 (70/20/10) after 77 replications (Figure 1). Fourteen of the sixteen portfolios are eliminated at the initial stage ($n_0 = 30$), leaving P04 and P01 (40/50/10) as survivors. P01 is then eliminated at stage 77, which leaves P04 as the final selection. Table 2 reports the extended evaluation over 200 CRN replications.

The cost decomposition in Figure 2 shows the mechanism behind this result. P04's adequacy penalty is much lower than that of its competitors (\$777M, compared with \$4,766M for P03 and \$7,205M for P09), while operating costs vary by less than 3% across the top-4 portfolios. The dominant source of differentiation is therefore reliability rather than dispatch efficiency. The physical driver is the interaction between low firm capacity and temporally correlated wind droughts produced by the NORTA AR(1) model. These events do not appear under iid sampling or mean-input evaluation, and they are responsible for the heavy right tails visible in Figure 8 for P03 and P09.

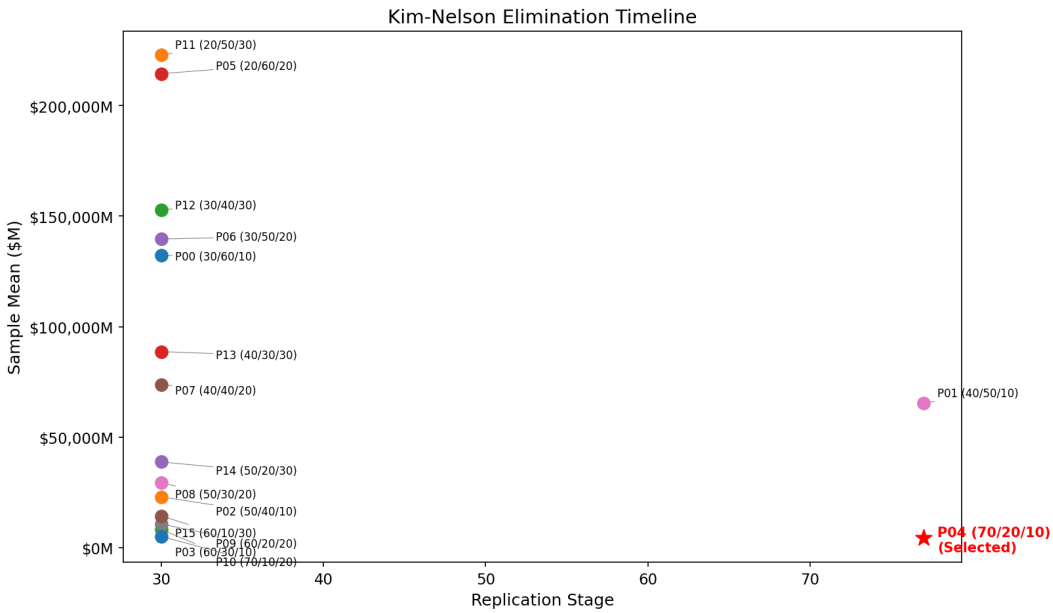

Figure 1: KN elimination timeline. Fourteen portfolios are eliminated at $n_0 = 30$; P01 is eliminated at stage 77, leaving P04 (70/20/10) as the selected portfolio.

Table 2: Top-4 portfolio evaluation (200 replications). RHW = relative half-width of the 95% confidence interval for mean Y.

Portfolio	Mean Y ($M)	F_i ($M)	$\bar{C}$ ($M)	$\lambda\bar{U}$ ($M)	RHW
P04 (70/20/10)	4,306	1,694	1,835	777	0.38%
P10 (70/10/20)	5,060	1,732	1,874	1,454	0.40%
P03 (60/30/10)	8,272	1,660	1,846	4,766	0.83%
P09 (60/20/20)	10,785	1,698	1,882	7,205	0.64%

All six pairwise differences among the top-4 portfolios are statistically significant under Bonferroni correction ($\alpha_{\text{pair}} = 0.0083$), and all relative half-widths are below 1% (Table 2). KN terminates at 77 replications, well below the 500-replication cap. This reflects the variance reduction that CRN achieves on pairwise differences (Figure 9).

5.2 Deterministic Benchmark Gap

Under deterministic assumptions (demand noise set to zero, all generators available, renewable capacity factors at their annual means), the lowest-cost portfolio is P03 (60/30/10) at $3,503M. This differs from the stochastic recommendation (P04). The DBG is:

$$\text{DBG} = \bar{Y}_{\text{P03}} - \bar{Y}_{\text{P04}} = \$3{,}968\text{M}, \quad 95\% \text{ CI: } [\$3{,}905,\ \$4{,}031]\,\text{M}. \tag{4}$$

A planner relying on mean-input evaluation would select a portfolio costing approximately $4B more per year when uncertainty is included (Figure 3).

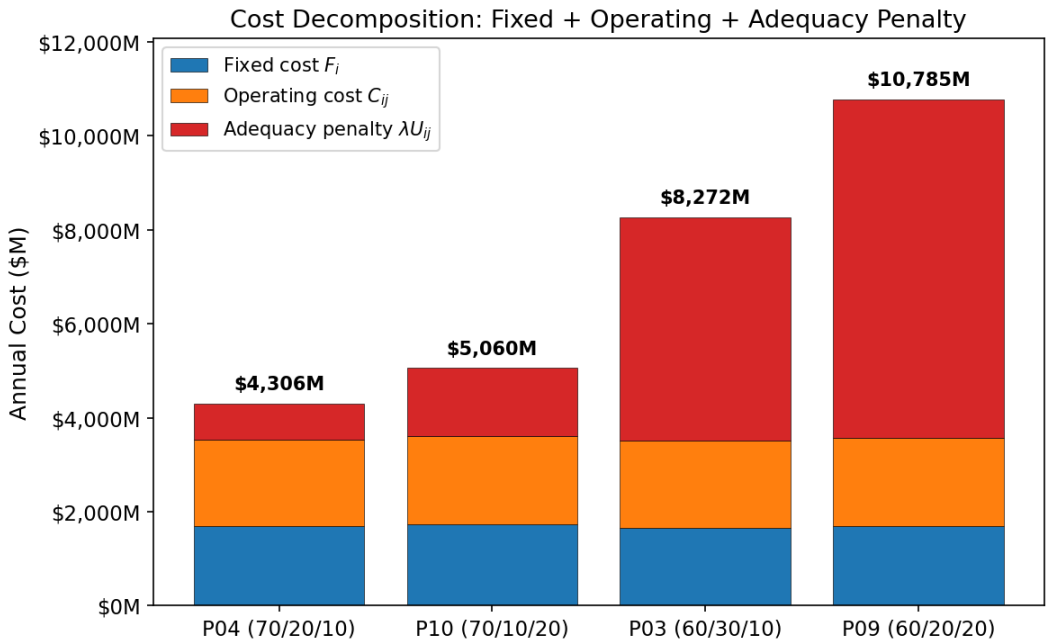

Figure 2: Cost decomposition: fixed cost F_i (blue), operating cost C_{ij} (orange), and adequacy penalty λU_{ij} (red). P04 pays the highest fixed cost but the lowest penalty, yielding the lowest total.

The source of the gap is informative. P03 saves \$34M in annualized fixed cost relative to P04 because its lower firm fraction requires less capital-intensive generation. This saving is only realized under mean-input evaluation, where renewables always meet their expected contribution and the 10 percentage-point firm deficit is invisible. Under stochastic evaluation the tail risk reappears. Demand noise ($\sigma_d = 0.09$) occasionally pushes load above the level that P03's firm fleet can cover, and forced outages (5–8% FOR) reduce available firm capacity precisely when it is needed most. The NORTA-modelled wind droughts compound this exposure. During multi-hour low-wind events, P03's 3,600 MW of renewables contribute negligibly, while its 7,200 MW of firm generation (reduced by outages) cannot bridge the gap. The \$4B DBG therefore quantifies the cost of ignoring these correlated tail events in portfolio selection. This finding directly answers Research Question 1. The preferred portfolio changes when uncertainty is included, and the cost consequence is substantial at the system scale.

5.3 Sensitivity Analysis

The 2^3 factorial (Figure 4) identifies demand noise (\$2,883M main effect) and forced outages (\$2,838M) as the dominant uncertainty factors. Renewable variability contributes only \$117M, reflecting the concentrated Beta(6.73, 15.69) marginal calibrated from fleet-level CODERS data: because aggregate Ontario wind capacity factors cluster tightly around 0.30, switching between stochastic and fixed-mean renewable profiles has limited impact on the performance gap between the best and worst portfolios.

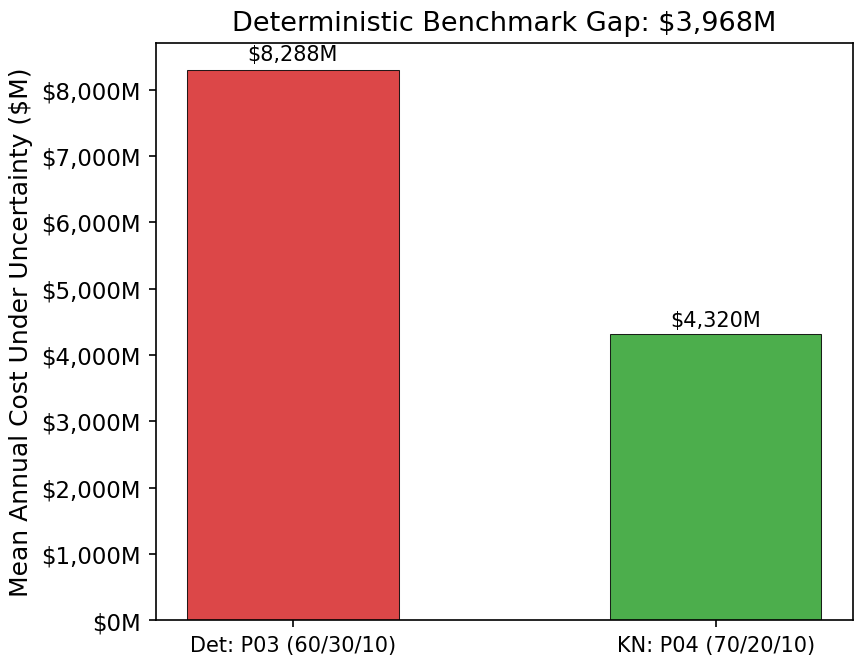

Figure 3: Deterministic Benchmark Gap. Under mean-input assumptions, P03 appears cheaper. Under stochastic evaluation with 200 paired CRN replications, P04 costs $3,968M less per year.

The negative demand×outage interaction (−$699M) reflects a saturation effect. When either factor alone is set to its high level, the most vulnerable hours are already experiencing shortfalls, so adding the second stressor has a diminishing marginal impact. Modelling demand uncertainty alone therefore captures much of the adequacy risk that outages would additionally reveal, and the same holds in reverse. The most important finding from the factorial is that P04 is the best portfolio in all eight configurations. This directly answers Research Question 2. Demand noise and forced outages move the *magnitude* of portfolio costs, but they do not change the *ranking*.

5.4 VOLL Sensitivity

To test robustness to the adequacy penalty coefficient, the top-4 portfolios were re-ranked for $\lambda \in \{5{,}000, 10{,}000, 15{,}000\}$ $/MWh using the same replication-level (F, C, U) outputs from the 200-replication CRN experiment. The selected portfolio remains unchanged (P04) across all tested values. This indicates the recommendation is not an artifact of a single VOLL calibration point, at least within a practically relevant range.

5.5 Adequacy Search

When total capacity is a decision variable, P04's mix achieves adequacy (EUE ≈ 0) at 15.8 GW, while P03 requires 17.5 GW and P09 requires 18.2 GW. The cheapest adequate portfolio is P04 at $3,763M.

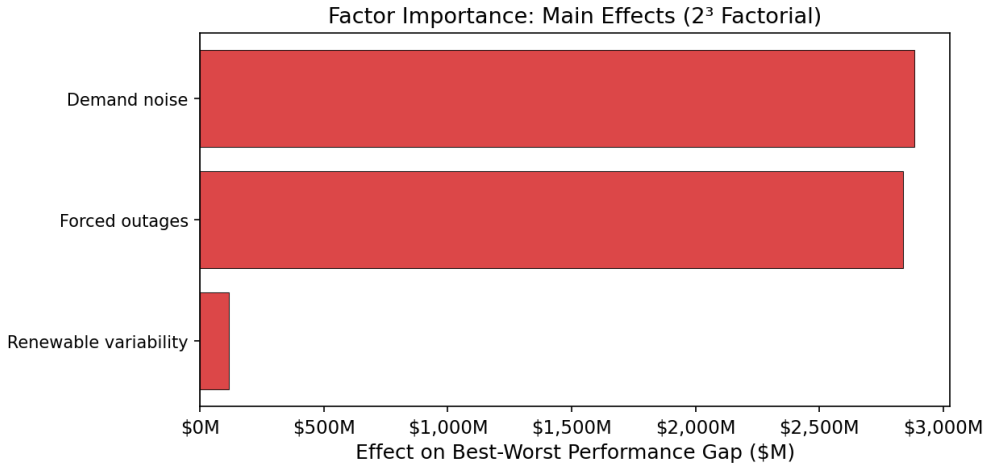

Figure 4: Factorial sensitivity analysis. Demand noise and forced outages dominate; renewable variability has minimal impact. P04 is the best portfolio in all eight configurations.

5.6 UCAP-Normalized Comparison and Empirical ELCC

To test whether equal-installed-MW comparisons bias the ranking toward high-firm portfolios, six mixes spanning 20% to 70% firm are evaluated at equal UCAP (8,000 MW). Even after normalization, P04 remains cheapest ($9,186M versus $10,167M for P03). This happens because static ELCC is a peak-period metric that does not capture sustained low-renewable events (Appendix, Figure 10).

To quantify the mis-pricing, an empirical ELCC experiment (Appendix H) measures the unserved-energy reduction from adding 1,000 MW of each technology to a stressed 5,000 MW firm baseline, and normalizes against a perfect-firm reference. The empirical values are **wind 0.25** (vs. 0.18 static), **solar 0.07** (vs. 0.35), and **storage 0.13** (vs. 0.50). The static ELCC values used in UCAP normalization therefore overstate the adequacy contribution of solar and storage by a factor of 4 to 5, which is why equal-UCAP portfolios still favour firm capacity.

5.7 OCBA Sanity Check and Capital Cost Sensitivity

Two robustness checks close the results section. First, an independent-sampling OCBA procedure [1] is run against KN as a sanity check on the 16-portfolio problem across 5 macro-replications at different base seeds. OCBA selects P04 in all 5 runs, matching the KN result (Appendix I). This is a consistency check on the KN selection rather than a powered comparison of the two procedures. Five macro-reps are too few for a meaningful $P(\text{CS})$ estimate, and the $750M gap between P04 and the runner-up makes this instance easy for either procedure.

Second, and more substantively, a ±30% one-at-a-time sensitivity on every capital-cost parameter (nuclear, flex firm, mid-merit, peaker, wind, solar, storage; sources in Appendix A) leaves P04 as the best portfolio in all 35 perturbed scenarios. The minimum P04-vs.-P10 gap across the entire sweep is $700M (Appendix G), well above the $50M indifference zone, so the central conclusion

is insensitive to plausible capital-cost calibration error. A verification-and-validation suite (Appendix F) independently confirms the simulator's deterministic limit, input moments, outage rates, monotonicity, and CRN variance reduction.

6 Conclusion

This study presents a simulation-based framework for comparing capacity portfolios under uncertainty. It combines chronological Monte Carlo simulation with empirically calibrated inputs, Common Random Numbers, and the Kim–Nelson ranking and selection procedure.

The central finding is that uncertainty changes the preferred portfolio. A deterministic analysis selects P03 (60/30/10), while the stochastic analysis selects P04 (70/20/10). P04 costs \$4B less per year when uncertainty is included (DBG = \$3,968M, $p < 0.05$). This result is robust across all eight factorial configurations and persists under UCAP normalization.

Three methodological contributions merit emphasis. First, the NORTA AR(1) input model preserves both the empirical marginal and the temporal autocorrelation ($\rho = 0.995$) of Ontario wind data. This matters because iid sampling underestimates sustained low-output events. Second, CRN achieves between 2× and 35× variance reduction on pairwise differences, which substantially accelerates KN elimination. Third, the adequacy search demonstrates stochastic root-finding for minimum reliable capacity, which connects portfolio evaluation to expansion planning.

Together, these results address both research questions. Research Question 1 is answered by the deterministic benchmark gap. The preferred portfolio changes from P03 under deterministic evaluation to P04 under stochastic evaluation, at a cost of \$3,968M per year if uncertainty is ignored. Research Question 2 is answered by the factorial analysis. Demand noise and forced outages are the dominant uncertainty factors by main-effect magnitude, but no tested factor combination changes the identity of the best portfolio. The recommendation is therefore robust to uncertainty calibration within the tested range.

These findings are consistent with the central insight of Park and Baldick [6], who show that incorporating uncertainty into capacity planning can change the preferred resource mix. Their study establishes this for expansion-timing decisions in a multi-stage stochastic program. The present study establishes the analogous result for portfolio ranking under a fixed-budget constraint. The methodological difference is consequential. Park and Baldick jointly optimize build decisions and uncertainty realizations, whereas this study takes candidate portfolios as given and applies formal statistical selection (KN with CRN) to identify the best performer. The shared conclusion is that deterministic planning can systematically undervalue firm capacity, because it fails to account for the tail risk introduced by correlated renewable shortfalls and generator outages.

Several simplifications in the simulation model are worth noting. Demand noise is iid, which may overstate the frequency of isolated spikes while understating sustained peak-demand episodes. Storage dispatch follows a rule-based heuristic rather than co-optimization across the full horizon. The NORTA AR(1) model assumes that the normal-space ρ equals the target autocorrelation, and

although the induced error is negligible at $\rho = 0.995$, it becomes relevant for lower autocorrelations. Forced outages are represented at a discrete 10-unit granularity per sub-type, which is coarser than a continuous-time Markov model. These simplifications are acknowledged as limitations, but none of them would reverse the central finding within the range of parameters tested.

The natural extension is a full stochastic capacity expansion formulation, which would jointly optimize the portfolio mix and the total capacity subject to an adequacy constraint. The adequacy search in this study solves a one-dimensional version of that problem, and the multi-dimensional formulation connects directly to the framework of Park and Baldick [6].

References

[1] C.-H. Chen, J. Lin, E. Yücesan, and S. E. Chick. Simulation budget allocation for further enhancing the efficiency of ordinal optimization. *Discrete Event Dynamic Systems*, 10(3):251–270, 2000. doi: 10.1023/A:1008349927281.

[2] Energy Modelling Hub. CODERS: Canadian open-source database for energy research and systems-modelling. `https://cme-emh.ca/en/coders/`, 2025. Release October 2025. Hourly provincial generation, demand, and capacity factor data.

[3] Independent Electricity System Operator. 2025 annual planning outlook. Technical report, Independent Electricity System Operator (IESO), Apr. 2025. URL `https://ieso.ca/-/media/Files/IESO/Document-Library/planning-forecasts/apo/2025/2025-Annual-Planning-Outlook.pdf`.

[4] S.-H. Kim and B. L. Nelson. A fully sequential procedure for indifference-zone selection in simulation. *ACM Transactions on Modeling and Computer Simulation*, 11(3):251–273, 2001. doi: 10.1145/502109.502111.

[5] B. D. Leibowicz, N. Zhang, J. P. Carvallo, P. H. Larsen, T. Carr, and S. Baik. The importance of capturing power system operational details in resource adequacy assessments. *Electric Power Systems Research*, 228:110057, 2024. doi: 10.1016/j.epsr.2023.110057.

[6] H. Park and R. Baldick. Multi-year stochastic generation capacity expansion planning under environmental energy policy. *Applied Energy*, 183:737–745, 2016. doi: 10.1016/j.apenergy.2016.08.164.

[7] L. A. Roald, D. Pozo, A. Papavasiliou, D. K. Molzahn, J. Kazempour, and A. J. Conejo. Power systems optimization under uncertainty: A review of methods and applications. *Electric Power Systems Research*, 214:108725, 2023. doi: 10.1016/j.epsr.2022.108725.

[8] O. Stover, P. Karve, and S. Mahadevan. Reliability and risk metrics to assess operational adequacy and flexibility of power grids. *Reliability Engineering & System Safety*, 231:109018, 2023. doi: 10.1016/j.ress.2022.109018.

[9] R. Wang, S. Wang, G. Geng, and Q. Jiang. Multi-time-scale capacity credit assessment of renewable and energy storage considering complex operational time series. *Applied Energy*, 355:122382, 2024. doi: 10.1016/j.apenergy.2023.122382.

A Consolidated Parameters

Table 3: Simulation parameters.

Parameter	Value	Basis
System		
Peak demand	8,000 MW	Stylized
Reserve margin	50%	Accounts for renewable ELCC
Total capacity	12,000 MW	Peak $\times$ 1.5
Demand		
Profile	2-harmonic OLS	IESO 2024 ($R^2 = 0.59$)
Noise σ_d	0.09	Residual std from fit
Renewables		
Wind CF	Beta(6.73, 15.69)	MLE, CODERS Ontario
Wind ρ	0.995	Lag-1 ACF, CODERS
Solar cloud	Beta(1.77, 6.91)	MLE, CODERS Ontario
Solar ρ	0.928	Lag-1 ACF, CODERS
Daylight hours	07:00–21:00	CODERS (UTC−5)
Wind/solar split	60/40	Stylized
Storage		
Duration	4 hours	Industry standard
Round-trip eff.	85%	NREL ATB
Discharge/charge	70th/30th pctile	Stylized rules
Cost		
VOLL (λ)	\$10,000/MWh	Mid-range
Nuclear capital	\$400/kW-yr	High overnight, 30-yr
Flex firm capital	\$100/kW-yr	Moderate
Mid-merit capital	\$70/kW-yr	Combined cycle
Peaker capital	\$70/kW-yr	Simple cycle
Wind capital	\$130/kW-yr	Onshore
Solar capital	\$100/kW-yr	Utility-scale
Storage capital	\$150/kW-yr	Battery, 15-yr life
KN procedure		
α / δ / n_0	0.05 / \$50M / 30	

B Portfolio Definitions

Table 4: All 16 candidate portfolios (12,000 MW total).

ID	Label	Firm	Renew	Storage	F_i ($M)
P00	30/60/10	3,600	7,200	1,200	1,557
P01	40/50/10	4,800	6,000	1,200	1,591
P02	50/40/10	6,000	4,800	1,200	1,625
P03	60/30/10	7,200	3,600	1,200	1,660
P04	70/20/10	8,400	2,400	1,200	1,694
P05	20/60/20	2,400	7,200	2,400	1,561
P06	30/50/20	3,600	6,000	2,400	1,595
P07	40/40/20	4,800	4,800	2,400	1,630
P08	50/30/20	6,000	3,600	2,400	1,664
P09	60/20/20	7,200	2,400	2,400	1,698
P10	70/10/20	8,400	1,200	2,400	1,732
P11	20/50/30	2,400	6,000	3,600	1,600
P12	30/40/30	3,600	4,800	3,600	1,634
P13	40/30/30	4,800	3,600	3,600	1,668
P14	50/20/30	6,000	2,400	3,600	1,702
P15	60/10/30	7,200	1,200	3,600	1,736

C ELCC Values

Table 5: ELCC values for UCAP normalization.

Technology	ELCC	Basis
Nuclear	94%	High availability baseload
Flexible firm	92%	Gas/hydro, slight derates
Mid-merit gas	95%	CCGT
Gas peaker	92%	Cycling stress
Wind	18%	IESO capacity auction
Solar	35%	Summer peak contribution
Storage (4h)	50%	Duration-limited

D Input Fitting Diagnostics

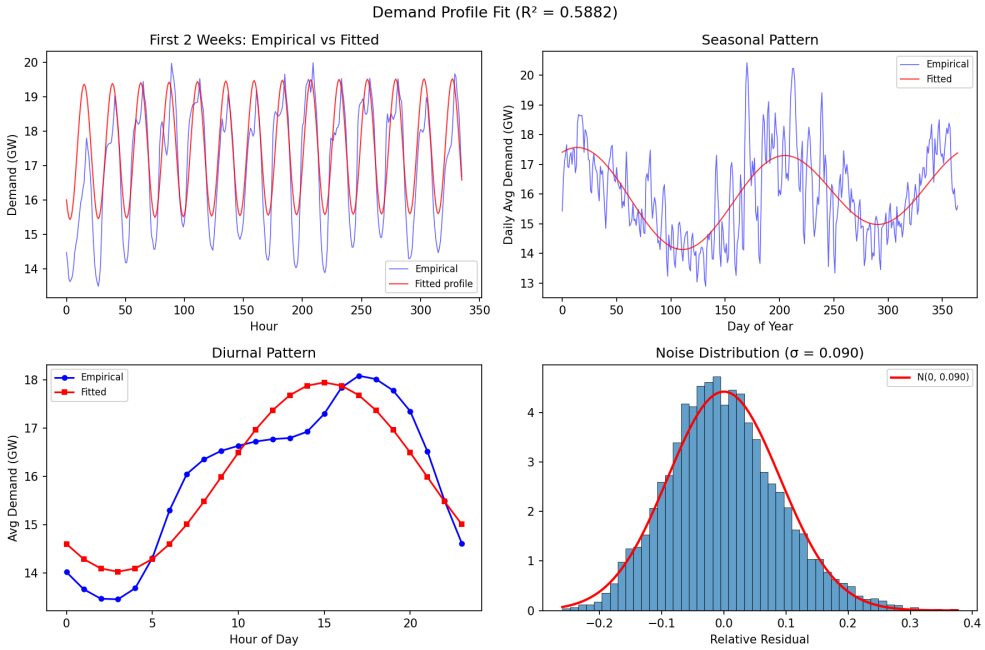

Figure 5: Demand profile OLS fit to IESO 2024 hourly Ontario demand. Left: observed vs. fitted. Right: residual distribution ($\sigma_d = 0.09$).

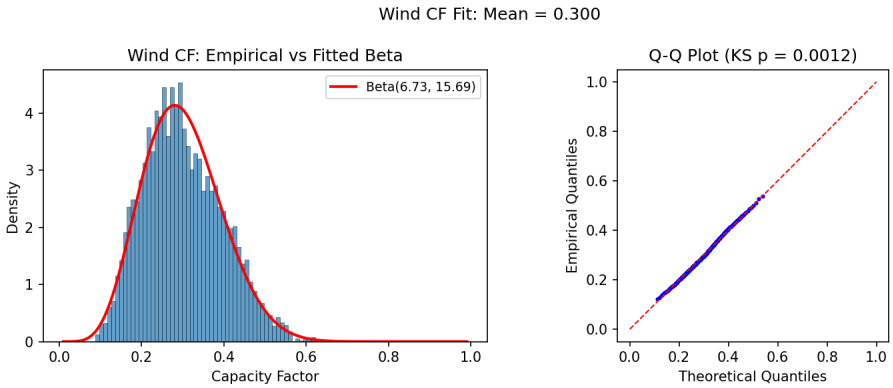

Figure 6: Wind CF Beta(6.73, 15.69) MLE fit to CODERS Ontario. Left: empirical vs. fitted PDF. Right: Q–Q plot (KS = 0.021).

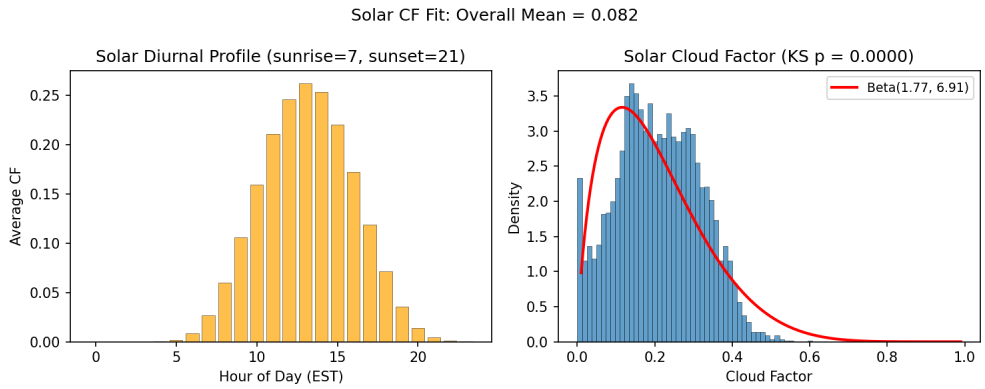

Figure 7: Solar CF: diurnal envelope × Beta(1.77, 6.91), calibrated from CODERS Ontario.

E Supplementary Results

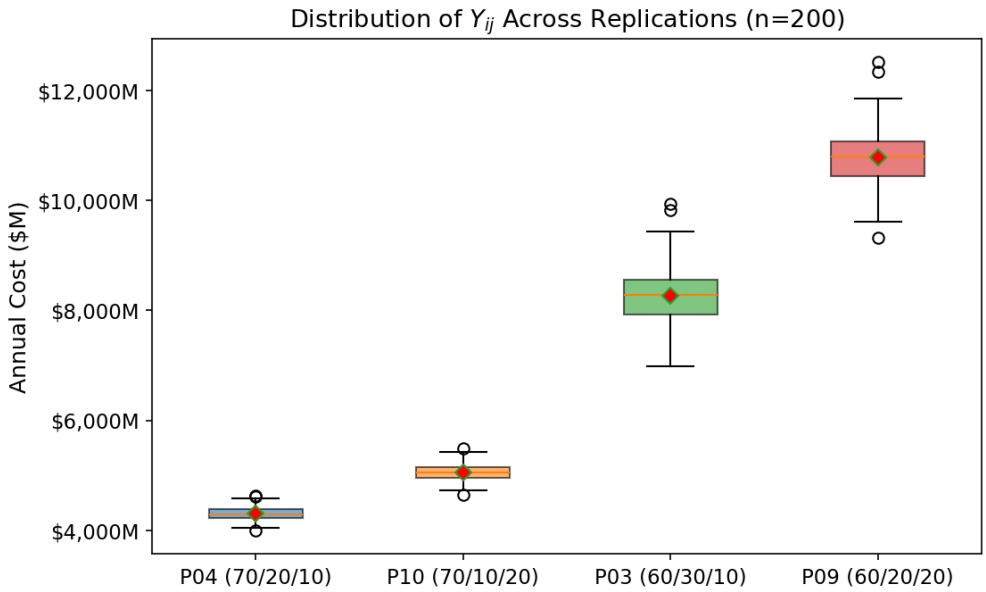

Figure 8: Distribution of Y_{ij} for top-4 portfolios. P03 and P09 exhibit heavy right tails from high-penalty replications during sustained low-renewable periods.

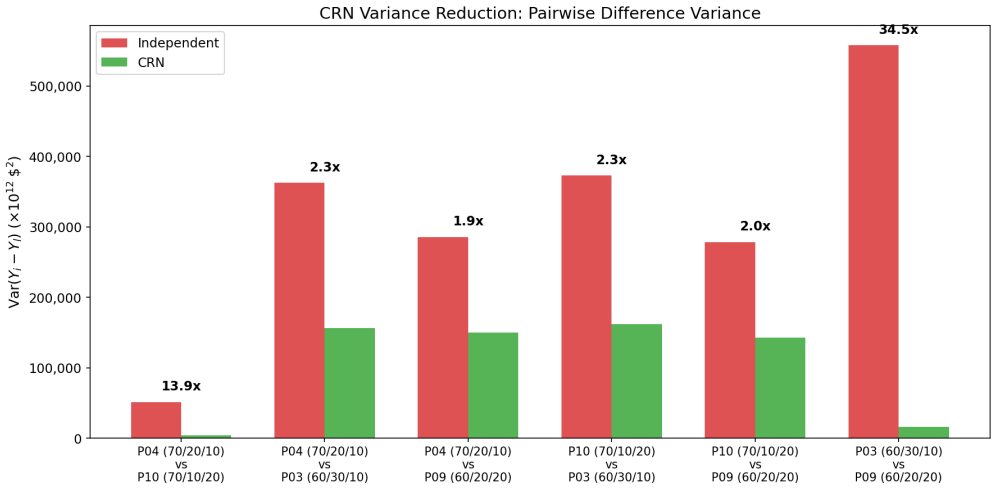

Figure 9: CRN variance reduction: pairwise difference variance under independent vs. CRN sampling. Efficiency ranges from 2× (P04 vs. P09) to 35× (P03 vs. P09).

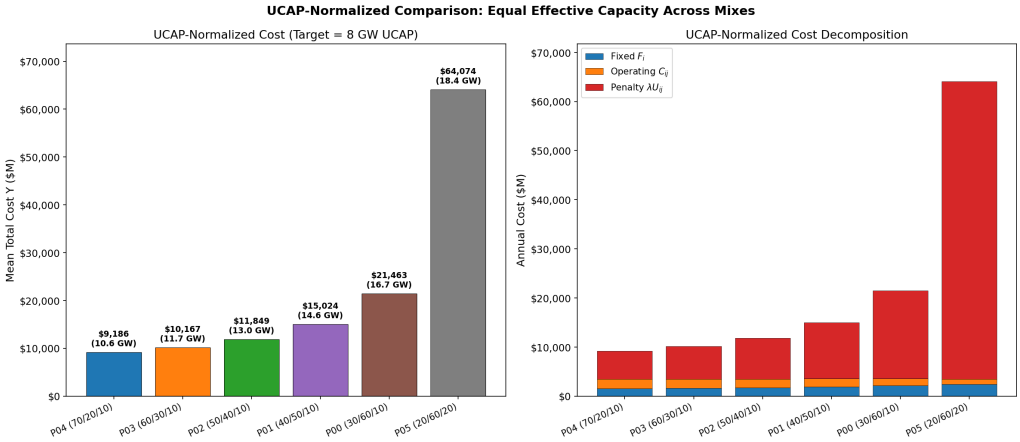

Figure 10: UCAP-normalized comparison at equal effective capacity (8 GW UCAP). Left: mean total cost per mix; right: cost decomposition. P04 remains cheapest after normalization.

F Verification and Validation

`src/verification.py`. Five automated checks, all pass.

Check	Observed (simulator)	Expected (analytical)	Pass
Deterministic limit of P04 (zero noise, zero FOR)	$U_{ij} = 0$; $Y_{ij} = \$3{,}505\text{M}$	$U = 0$; $Y = F + C_{\text{det}}$	✓
Input mean recovery (50 reps)	demand 6,549 MW (<0.01%); wind CF 0.304 (1.2%)	demand 6,549 MW; Beta marginal 0.300	✓
Outage-rate recovery (50 reps, all four sub-types)	within 0.1 pp of $1 - \text{FOR}$	$1 - \text{FOR}$	✓
Monotonicity: low-firm vs. high-firm EUE	P05: 21.2M MWh P04: 0.07M MWh (285×)	$\text{EUE}_{\text{low firm}} > \text{EUE}_{\text{high firm}}$	✓
CRN variance reduction on P04 − P10	sd = \$63.7M (8.8× reduction)	< independent sd = \$189.1M	✓

The rule-based storage and discrete 10-unit outage model (Section 6) are reported limitations rather than V&V failures, because neither has an analytical target to verify against.

G Capital Cost Sensitivity

`src/capital_sensitivity.py`. One-at-a-time ±30% sweep on each of the seven capital-cost parameters (Appendix A); F_i recomputed, C_{ij} and U_{ij} held fixed from the 200-replication CRN baseline. P04 remains best in all 35 perturbed configurations; minimum P10−P04 Mean Y gap = \$700M (storage +30%), well above the \$50M indifference zone.

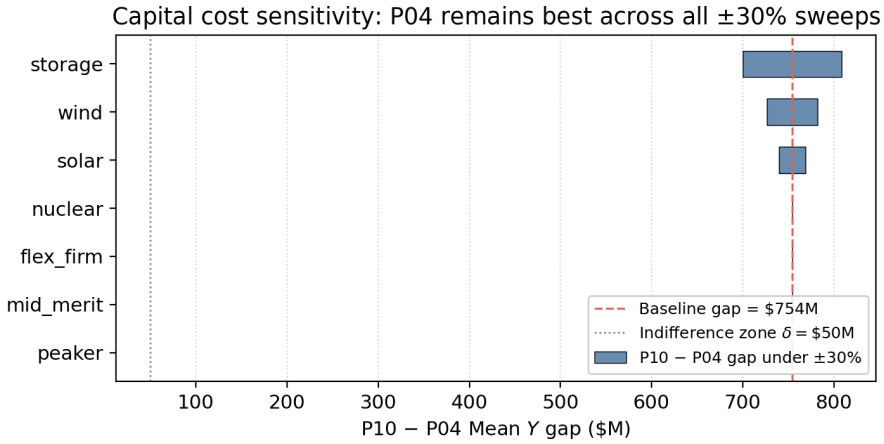

Figure 11: P10−P04 Mean Y gap under ±30% one-at-a-time perturbations of each capital cost parameter. Dashed red: baseline gap. Dotted grey: \$50M indifference zone.

H Empirical ELCC Experiment

`src/empirical_elcc.py`. Method: add a 1,000 MW dose of technology X to a stressed 5,000 MW firm-only baseline S_0 (realistic FOR, no renewables, no storage), measure $\mathbb{E}[U_X]$ over 100 CRN

replications, and define $\mathrm{ELCC}_X^{\mathrm{emp}} = (\mathbb{E}[U_0] - \mathbb{E}[U_X]) \,/\, (\mathbb{E}[U_0] - \mathbb{E}[U_{\mathrm{perf}}])$, where U_{perf} is obtained by adding 1,000 MW of firm with all FORs overridden to zero. Baseline $\mathbb{E}[U_0] = 16.14$M MWh; perfect-firm reference reduction $\Delta U_{\mathrm{perf}} = 9.98$M MWh.

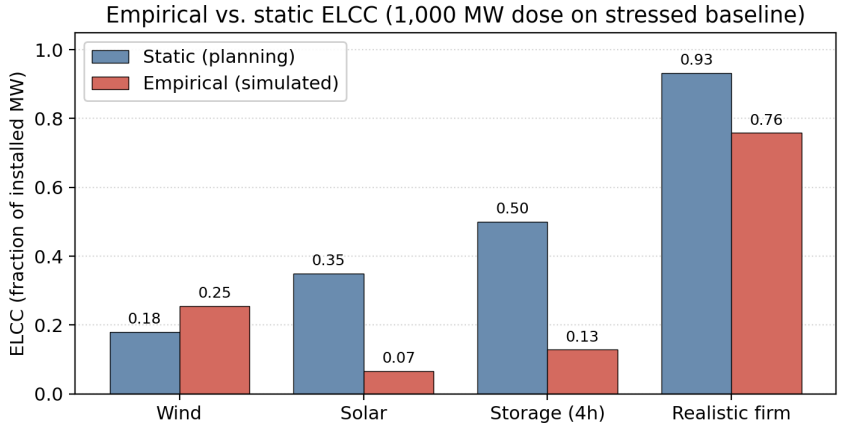

Figure 12: Empirical vs. static ELCC, 1,000 MW dose on the 5,000 MW stressed firm baseline. Solar and storage come in at ∼1/5 and ∼1/4 of their static planning values.

I OCBA Sanity Check

`src/ocba.py`. OCBA [1] with classical independent sampling is run against KN (with CRN) across five macro-replications at different base seeds. The purpose is a consistency check on the KN selection, not a powered comparison of the two procedures. Five macro-reps are too few for a meaningful $P(\mathrm{CS})$ estimate, and the ≈\$750M gap between P04 and the runner-up makes the selection easy for either method. In each macro-rep, KN terminates naturally and OCBA is given a matching simulation-call budget.

Table 6: OCBA sanity check (5 macro-reps, $k = 16$, $n_0 = 30$). Ground-truth best P04 from the baseline KN run.

Procedure	Selected P04	Mean calls	Samples on P04 (avg)
KN (with CRN)	5/5	528	54.0 (10.2%)
OCBA (independent)	5/5	528	78.0 (14.8%)

Both procedures select P04 in every run on this instance. This is consistent with the KN result but does not distinguish the methods; the allocation profiles differ sharply (Figure 13), which is qualitative evidence of OCBA's concentration behaviour but not a performance claim.

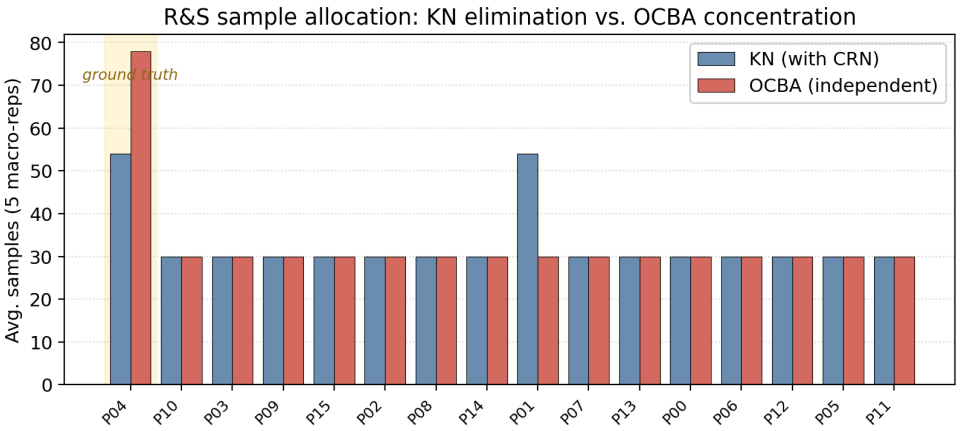

Figure 13: Average sample allocation per portfolio (5 macro-reps), ordered by baseline Mean Y. KN elevates both finalists (P04, P01) to 54 samples each; OCBA places 78 on P04 alone.

J Code Architecture and AI Disclosure

Python 3.11 (NumPy, SciPy); sixteen source files totalling ∼3,000 lines. Full pipeline runs in ∼5 min on a consumer laptop; each extension (`verification`, `capital_sensitivity`, `empirical_elcc`, `ocba`) is a separately runnable module. Base seed 42 for the core pipeline; seeds 100–104 for OCBA macro-reps. Reproduce via `python src/main.py` and `python src/<module>.py` for each extension.

Generative AI (Claude, Anthropic) was used for code development, debugging, statistical methodology review, and drafting support. All modelling decisions, parameter choices, experimental design, interpretation, and final writing are the author's own.